

KETHA BHANU TEJA

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with a B.Tech in Computer Science & Engineering (AI & ML). Demonstrated expertise in building end-to-end Machine Learning pipelines, Deep Learning (CNNs), and Retrieval-Augmented Generation (RAG) systems using Python, TensorFlow, PyTorch, and LangChain. Proven track record of boosting model accuracy up to 92% and optimizing real-time data pipelines in agile engineering teams.

CORE TECHNICAL SKILLS

Languages & Core CS: Python, SQL, JavaScript, HTML5, CSS3, Data Structures & Algorithms, Object-Oriented Programming (OOP)

AI / ML & Frameworks: TensorFlow, PyTorch, Keras, Scikit-Learn, LangChain, Hugging Face, OpenCV, FAISS, Pandas, NumPy, Streamlit

Developer Tools & Databases: MySQL, Git, GitHub, Jupyter Notebook, Visual Studio Code, MS Excel

Domain Expertise: Retrieval-Augmented Generation (RAG), Convolutional Neural Networks (CNNs), SMOTE Data Imbalance, Prompt Engineering

EXPERIENCE & INTERNSHIPS

AI-ML Virtual Intern

Oct 2024 – Dec 2024

AICTE – EduSkills

Virtual

- Engineered and deployed 6 end-to-end Machine Learning pipelines using Python, Scikit-Learn, and TensorFlow, increasing model accuracy by 12%.
- Mitigated severe class imbalance across multi-class datasets using SMOTE, reducing false-negative rates by 9% and enhancing precision-recall.
- Standardized feature preprocessing and data transformation workflows alongside senior engineers, accelerating pipeline deployment cycle by 20%.

Machine Learning Research Intern

Aug 2023 – May 2024

IIIT Hyderabad (Foundation of Modern Machine Learning)

Hyderabad, AP

- Architected and evaluated 6 machine learning model variants on a 10,000+ record dataset via hyperparameter tuning, achieving 0.89 peak ROC-AUC.
- Accelerated production deployment decisions by documenting model performance trade-offs alongside research leads for final deployment selection.
- Automated benchmark testing and performance logging across experimental neural networks using Python and Pandas, cutting evaluation time by 25%.

PROJECTS

AI-Powered Multi-Document RAG Chatbot

Jun 2025 – Aug 2025

Tech Stack: Python, LangChain, FAISS, Streamlit, Hugging Face Embeddings, OpenAI GPT

- Architected an enterprise RAG chatbot combining LangChain, FAISS vector index, and OpenAI GPT for context-aware Q&A across PDF repositories.
- Optimized document ingestion pipelines via custom chunking and Hugging Face embeddings, improving retrieval accuracy by 35% and lowering latency.
- Designed an interactive full-stack web UI using Streamlit, enabling real-time document parsing and context-aware multi-turn conversations.

Real-Time Facial Expression & Emotion Detection System (CNN)

Jan 2025 – Mar 2025

Tech Stack: Python, TensorFlow/Keras, OpenCV, Streamlit

- Engineered a real-time computer vision emotion recognition system using Deep CNNs and OpenCV, achieving 92% validation accuracy.
- Reduced model overfitting by 40% by implementing data augmentation, dropout regularization, and batch normalization during model training.
- Deployed a live Streamlit web application with webcam streaming integration, processing frames at 30+ FPS for low-latency facial prediction.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI & ML)

Nov 2021 – May 2025

Kakinada Institute of Engineering and Technology (JNTUK), Korangi, AP

CGPA: 7.20 / 10.0

Intermediate (MPC - Mathematics, Physics, Chemistry)

Jun 2019 – Apr 2021

Aditya Junior College (BIEAP), Mandapeta, AP

CGPA: 6.6 / 10.0

Secondary School Certificate (SSC)

Jun 2018 – Apr 2019

ZP High School (BSEAP), Ubalanka, AP

CGPA: 9.7 / 10.0

Certifications & Achievements: Python Foundation (Infosys Springboard) • Fundamentals of ML (IIIT Hyderabad) • Presented ML Research Papers at National Symposia • 1st Place Champion, Inter-College Kabaddi Tournament.